

Opinion

The brain bases of emotion generation and emotion regulation

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Emotion generation and emotion regulation are widely seen as functionally distinct. This distinction has inspired efforts to define separable brain bases of each, with emotion generation thought to involve mainly subcortical structures such as the amygdala, and emotion regulation thought to involve mainly cortical regions such as fronto-parietal cortices. However, emerging findings challenge strong neural separability accounts, revealing substantial overlap between brain systems underlying emotion generation and emotion regulation. In this opinion article we review evidence that supports and challenges neural separability. Based on this evidence, we propose an updated neural account using a valuation-systems perspective that incorporates both common and distinct brain bases of emotion generation and emotion regulation. Finally, we discuss implications for next-generation study designs, methods, and clinical interventions.

The distinction between emotion generation and emotion regulation

One of the most fundamental distinctions in affective science is the distinction between emotion generation and emotion regulation. This distinction is rooted in everyday experience, and is most obvious in social settings in which emotion-generative processes are operative in one person, and emotion-regulatory processes are operative in another person. For example, a parent may help soothe a child who is upset at the loss of a cherished toy, or a co-worker may provide a positive reframe for a colleague devastated at being passed over for promotion. However, the distinction also applies within a single individual, such as when someone becomes angry and then subsequently tries to control their anger. In each case, there seems to be a clear distinction between emotion generation and emotion regulation, where the latter is defined by the activation of a goal to regulate the emotion-generative process.

This distinction between emotion generation and emotion regulation is made across many different fields, including psychology [1], artificial intelligence [2], medicine [3], law [4], business [5], sociology [6], and anthropology [7]. Within psychology, researchers regularly reference this distinction in biological [8], developmental [9], social [10], personality [11], health [12], and clinical [13] areas.

Widespread acceptance of this basic distinction has inspired efforts to identify the brain bases of emotion generation and emotion regulation. One common assumption made over the last two decades is that the clear psychological distinction between emotion generation and emotion regulation should be paralleled by a clear neural distinction (e.g., [14–16]). However, this view has not been universally endorsed [17–19]. In what follows, we review the initial articulation of this neural separability assumption along with supporting evidence and then describe recent evidence that raises questions about it. We then provide an updated framework for advancing our understanding of the neural bases of emotion generation and emotion regulation and suggest future research directions.

Highlights

Emotion generation and emotion regulation are typically thought to engage distinct neural systems.

This neural separability assumption has contributed to a common, but erroneous, 'reverse' inference that prefrontal activation directly indicates emotion regulation.

Mounting evidence suggests that there is substantial overlap in the brain systems underlying emotion generation and emotion regulation.

We provide an updated neural account based on the process model of emotion regulation, which highlights the context-dependent role of common and distinct brain systems.

This framework may be helpful for fashioning brain-based interventions to improve emotion regulation.

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The neural separability assumption

An early articulation of the neural separability assumption postulated that emotion generation was encapsulated in subcortical structures whose activity was modified by regulatory systems encapsulated in neocortical regions [15,16]. This view leveraged theories of cognitive control in cognitive neuroscience [20] and extended them to the affective domain. According to this view, subcortical and some 'limbic' cortical regions mediate evolutionarily conserved affective responses, which are modulated by context- and goal-dependent activity in the prefrontal cortex (PFC) [20,21].

This approach has ample historical precedent, including Descartes' idea of reason versus the passions [22] and Hughlings Jackson's idea of the cortex as a controlling and inhibitory structure [23]. The motivating idea was that emotion-generative brain systems, including the amygdala, give rise to emotion, while emotion-regulatory brain systems, including the prefrontal and cingulate cortices, exert modulatory influences on the emotion-generative systems. Neural separability remains a widely held assumption, with fronto-parietal regions thought to be implicated in emotion regulation and various (largely subcortical) regions such as amygdala, ventral striatum (VS)/nucleus accumbens (NAcc), insula, and periaqueductal gray (PAG) thought to be implicated in emotion generation (Figure 1A) [14,24].

Temporally, these systems could interact with each other in an iterative fashion [25,26], but each plays a clearly separable role, allowing generation and regulation to be independently assessed

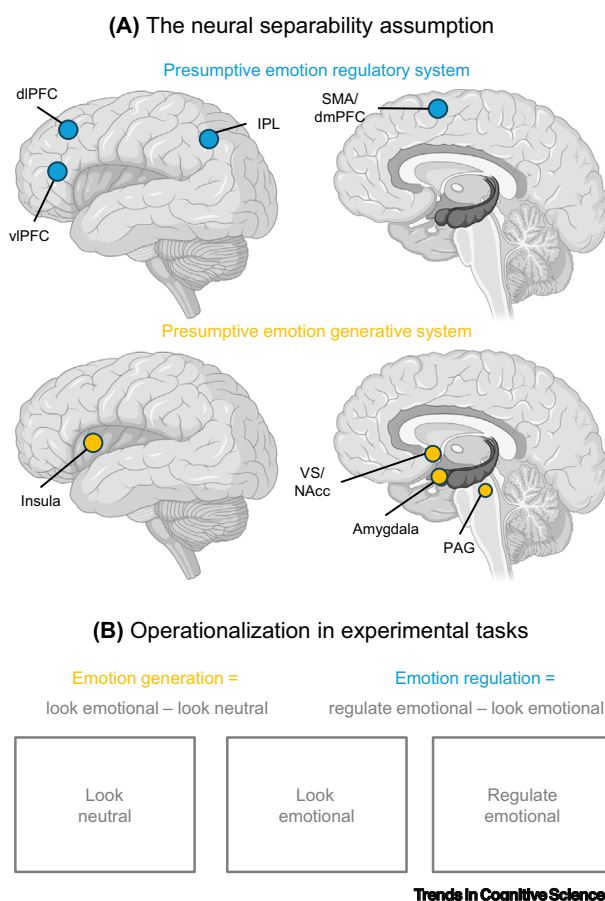


Figure 1. Emotion generation and emotion regulation. (A) The neural separability assumption. In this view, emotion generation and emotion regulation are subserved by separate but interactive systems. The emotion-regulatory system usually includes dorsal and ventral lateral prefrontal cortices (dIPFC and vIPFC), supplementary motor area (SMA), and inferior parietal lobe (IPL). The emotion-generative system usually includes the amygdala, ventral striatum (VS)/nucleus accumbens (NAcc), insula, and periaqueductal gray (PAG). (B) Operationalization of emotion generation and emotion regulation in experimental tasks. In a typical task assessing emotion generation and emotion regulation, there are three experimental conditions, a 'look-neutral' condition with viewing an emotionally neutral stimulus, a 'look-emotional' condition with viewing an emotionally negative or positive stimulus, and a 'regulate-emotional' condition with regulating emotion in the presence of an emotional stimulus. Emotion generation is generally operationalized as the contrast between the 'emotion' and 'neutral' conditions in behavioral or neural signals. Emotion regulation is generally operationalized as the contrast between the 'regulation' and 'emotion' conditions.

and manipulated. In this view, the emotion-generative system can be activated without engaging regulatory systems, leading to emotional states and behaviors. Activation of the regulatory system may or may not modulate the emotion-generative system, depending on the effectiveness of the regulation attempt. This hypothesis is well aligned with the commonsense psychological distinction between emotion generation and emotion regulation.

Evidence corroborating the neural separability assumption

This neural separability assumption attracted increasing attention, beginning in the early 2000s, coincident with the rise of functional magnetic resonance imaging (fMRI). Given its relatively good brain coverage and spatial resolution, researchers were motivated to search for brain regions relatively more activated during emotion regulation, which were presumed to be part of 'emotion regulation' brain systems, or relatively less activated during down-regulation of emotion, which were presumed to be part of 'emotion generation' brain systems.

Over the last two decades, the neural separability assumption has been tremendously popular. It has provided conceptual grounding for over 100 fMRI studies (over 300 experiments) [27], generated important findings in a range of populations, and advanced our understanding of the neural bases of emotion. Most studies have focused on reducing negative emotions (i.e., downregulation) using emotion-regulation strategies such as reappraisal and distraction [1]. Following early fMRI studies of emotion regulation [16,28], studies have found that emotion regulation (see operationalization in Figure 1B) is generally associated with activation (i.e., relative increase in neural signal compared with a baseline condition) in fronto-parietal regions, including the dorsal and ventral lateral PFC (dlPFC and vlPFC), supplementary motor area (SMA) extending to the dorsal medial PFC (dmPFC), and the inferior parietal lobe (IPL) (e.g., [29–37]) (Figure 1A). Among the studies of emotion downregulation, some (but not all) have also found reduced amygdala activation during emotion regulation (e.g., [16,29–32,34,35]). Consistently, emotion-regulation success has been correlated across individuals with activation in these fronto-parietal regions [16,33,34], deactivation (i.e., relative decrease in neural signal compared with a baseline condition) in amygdala [16,37], and activation in reward-related VS regions [34,38]. Furthermore, studies have also investigated neural activity during emotion regulation in different age groups (e.g., [39,40]), genders (e.g., [41,42]), and clinical populations (e.g., [43–45]), reporting largely the same fronto-parietal activations despite some quantitative differences between groups. A number of neuroimaging meta-analyses over the years have confirmed these general findings [27,46,47].

Other methods have provided converging results regarding neural separability. Scalp EEG studies have shown increased theta and gamma oscillations in frontal regions during emotion regulation [48,49]. One intracranial EEG study in epilepsy patients showed reduced event-related activity in amygdala and insula during emotion regulation [50]. Studies using event-related potentials (ERPs) have linked the late positive potential (LPP), an ERP component that emerges around 300 ms post-stimulus onset, to the generation of positive and negative emotion [51–53]. LPP responses to emotional stimuli are attenuated by emotion regulation strategies [54].

Another approach that has provided further insight is neurostimulation, including transcranial direct-current stimulation (tDCS) and transcranial magnetic stimulation (TMS), among others. These methods have been used to target specific brain regions in the hopes of enhancing emotion regulation and thereby decreasing negative emotion, with the ultimate goal of improving mental well-being and benefiting clinical populations. To date, most neurostimulation research is grounded in the neural separability assumption. As presumptive 'regulatory' regions, dlPFC and vlPFC are commonly targeted to enhance emotion regulation. In some studies, tDCS and

TMS stimulation on these regions has led to better outcomes during emotion regulation, such as greater reduction in negative affect elicited by unpleasant images and social pain [55–57].

Taken together, findings from these neuroimaging, neurophysiology, and neurostimulation studies seem to converge in suggesting that emotion regulation is associated with activation in fronto-parietal regions and (in some cases) deactivation in amygdala. Such findings have been widely seen as corroborating the neural separability assumption, and this has led to widespread reverse inferences about emotion regulation from neuroimaging results [58,59]: for example, the inference that increased frontal activity indicates emotion regulation (e.g., [60]). Consequently, clinical interventions have specifically targeted frontal regions in psychopathological conditions associated with emotion dysregulation [61,62]. For example, neurostimulation therapies targeting the PFC have been applied to patients with depression, bipolar disorder, anxiety disorder, and PTSD, albeit with only moderate clinical improvement compared with controls [63,64].

Evidence challenging the neural separability assumption

While the neural separability assumption is widespread, many findings do not fit neatly into this framework. For instance, a number of neuroimaging studies have found widespread and robust cortical activation during emotion generation (e.g., [65–67]) and a lack of robust amygdala deactivation during emotion downregulation [33,36,37]. In addition, in value-based decision-making studies, a putative ‘regulation’ region, namely dlPFC, has been shown to be involved in the construction of value rather than the inhibition of affective responses [68,69]. In these studies, dlPFC activity increases as value-related evidence is accumulated in service of both affective (i.e., hedonic) and regulatory goals. A similar role for the dlPFC has been demonstrated in evaluating the relative value of different choice options [70]. In the context of emotion generation and emotion regulation, this maps onto a general role of the dlPFC in affective appraisal, whether it is used to generate or regulate emotion.

Another type of puzzling evidence comes from neurostimulation studies targeting presumptive ‘regulation’ regions, such as the dlPFC and vlPFC. First, neurostimulation of these regions often produces null effects [71–73], despite some reports of successful enhancement of emotion regulation [55–57,74]. Second, some studies have found PFC neurostimulation effects on emotion generation instead of emotion regulation. For instance, one study found that dlPFC-targeted tDCS did not significantly affect emotion regulation but did reduce emotion reactivity in a subsequent task [71], and similar effects have been reported for the disruption of vlPFC using TMS [75]. These discrepancies invite reconsideration of the neural separability assumption.

Although the neural separability assumption has guided research for decades, it has rarely been rigorously tested as a formal hypothesis. Following the neural separability assumption (Figure 1A), we propose two predictions that are testable with fMRI imaging: (i) during emotion generation, emotion-generative regions should be activated, but emotion-regulatory regions should not be activated, and (ii) during emotion downregulation, emotion-regulatory regions should be activated, but emotion-generative regions should either remain unchanged or be deactivated. Two separate large-sample studies have tested these predictions. The first [76] directly compared the fMRI activation profiles associated with emotion regulation and emotion generation across presumptive ‘generation’ regions (e.g., amygdala and insula) and ‘regulation’ regions (e.g., frontal-parietal regions and SMA) using a sample of 336 participants from both healthy and clinical populations (Figure 2A). Following the neural separability assumption, one would expect greater activation for the generation contrast than the regulation contrast (see operationalization in Figure 1B) in ‘generation’ regions and the opposite pattern in ‘regulation’ regions. However, the researchers observed the same pattern (greater activation for generation than regulation) in both types of regions. Using multivariate pattern analysis (MVPA),

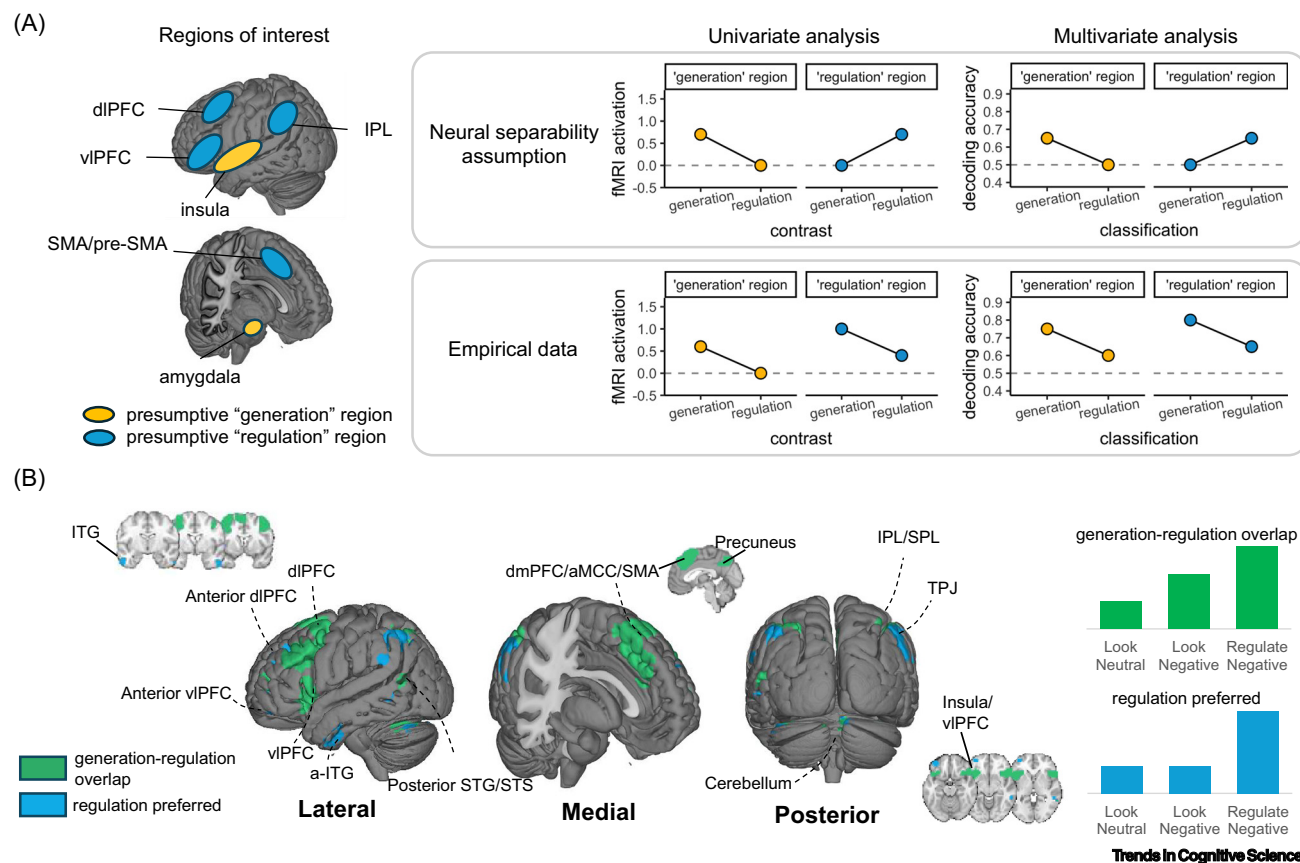


Figure 2. Tests of neural separability between emotion generation and emotion regulation using functional magnetic resonance imaging (fMRI). (A) A region-of-interest approach to test the neural separability assumption [76]. This study ($n = 336$) selected a small number of presumptive 'generation' regions (amygdala and insula) and 'regulation' regions (frontal-parietal regions and supplementary motor area, SMA) from the literature. In these regions, the research team compared the fMRI activation for the generation contrast (look negative > look neutral) and the regulation contrast (regulate negative > look negative) in univariate analysis, and the decoding accuracy in the generation classification between look-neutral versus look-negative and the regulation classification between look-negative versus regulate-negative in multivariate analysis. Contrary to the predictions of the neural separability assumption, the empirical data showed that both presumptive 'generation' and 'regulation' regions displayed greater activation for the generation contrast than for the regulation contrast, and greater decoding accuracy for the generation classification than the regulation classification. (B) A whole-brain approach to test the neural separability assumption [77]. This study ($n = 358$) used a system identification approach using Bayes factor analysis to identify common and unique brain regions for emotion generation and emotion regulation across the brain. The research team searched the brain for 'regulation preferred' regions, which show positive activation for regulation contrast and null activation for generation contrast, and 'generation-regulation overlap' regions, which show positive activation for both the generation contrast and the regulation contrast. Contrary to the neural separability assumption, they identified large areas across the brain as 'generation-regulation overlap' regions – including fronto-parietal regions, dorsal medial prefrontal cortex (dmPFC)/SMA/anterior midcingulate cortex (aMCC), and insula – while the 'regulation preferred' regions were identified in areas including anterior prefrontal cortices, posterior temporoparietal junction (TPJ), and anterior inferior temporal gyrus (ITG), comprising only 15% of the volume of the overlapping regions. Other abbreviations: dIPFC, dorsolateral prefrontal cortex; IPL, inferior parietal lobe; STG/STS, superior temporal gyrus/superior temporal sulcus; vIPFC, ventrolateral prefrontal cortex.

similarly, they found higher accuracy in the generation classification (between look-neutral and look-negative) than the regulation classification (between look-negative and regulate-negative) using fMRI data, whether from 'generation' or 'regulation' regions. Of note, compared with univariate analysis, MVPA uses data from multiple voxels simultaneously and returns a multivoxel pattern rather than an overall activation level. Thus, it yields more fine-grained information about the functional role(s) of a brain region than the univariate analyses that most prior studies have used. These results show that the presumptive 'regulation' regions tested are not selectively engaged in emotion regulation and thus challenge the neural separability assumption.

The second study (total $n = 358$) tested these predictions using a whole-brain method [77]. Here, the researchers used a system identification approach using Bayes factor analysis to identify common and unique brain regions for emotion generation and emotion regulation across the brain, without restricting the analysis to predefined regions (Figure 2B). This approach identified ‘overlap regions’ showing strong evidence for positive activation in both generation and regulation contrasts (see operationalization in Figure 1B), while ‘regulation-preferred’ regions showed strong evidence for null activation in the generation contrast but positive activation in the regulation contrast. These regions were activated when viewing negative images without an explicit regulatory goal, and further activated when downregulating negative emotion. Overlap regions included both presumptive ‘regulation’ regions (fronto-parietal regions, dmPFC/SMA/anterior midcingulate cortex (aMCC) and presumptive ‘generation’ regions (e.g., insula) (cf. Figure 1A). These results show a substantial overlap between emotion-regulatory and emotion-generative brain systems, challenging the neural separability assumption.

It is worth noting that while most brain regions show overlap between emotion regulation and emotion generation, this study [77] also identified a set of regions whose responses were consistent with the neural separability assumption, at least in the context of downregulation of responses to emotional images. Specifically, fMRI activation in anterior prefrontal cortices and posterior temporoparietal junction (TPJ) increased with regulatory demand (regulate-emotional versus look-emotional) but were unaffected by the affective content (look-negative versus look-neutral) of the stimuli. Conversely, several cortical and subcortical regions involved in emotion generation were unaffected by reappraisal, signaling a selective role in emotion generation. Together, these findings support a view in which emotion generation and emotion regulation systems are largely overlapping and only partially separable, but these findings may depend on the stimulus, strategy, practice, and population.

One potential concern regarding the apparent failure to find support for strong neural separability is that people might spontaneously regulate their emotions even without explicit instructions [78]. Such spontaneous regulation could explain why many presumptive ‘regulation’ regions appear to be involved in both emotion generation and emotion regulation. However, we suggest that spontaneous regulation alone is inadequate to explain this overlap, based on two key pieces of evidence. First, if spontaneous regulation were the primary driver of this shared activation, we would expect activation in overlapping regions to negatively correlate with participants’ negative emotion ratings during emotion generation. However, the first study [76] found that the correlation between activity in overlapping regions and negative emotion ratings was nearly zero, providing no support for the first prediction. Second, if spontaneous regulation was the driving mechanism, one would expect all regulation-related regions to be activated during emotion generation. However, the second study [77] identified ‘regulation-preferred’ regions that were not activated during emotion generation, violating the second prediction and suggesting that any potential spontaneous activity may not involve the same brain processes as the actual emotion-regulation system. Together, these findings suggest that while spontaneous regulation might play a role, it is unlikely to fully account for the involvement of these fronto-parietal regions during emotion generation.

An updated neural account

When considering the degree of overlap in the neural systems relevant to emotion generation versus emotion regulation, we find it useful to imagine a continuum of possible overlap, ranging from no overlap (i.e., full separability) on the left side of the continuum to full overlap (i.e., no separability) on the right (Figure 3A). Broadly speaking, various theoretical perspectives stand at different positions on this continuum [19]. On one side (left in Figure 3A), basic emotion perspectives [79] hold that the neural systems relevant to emotion generation and emotion regulation are distinct and

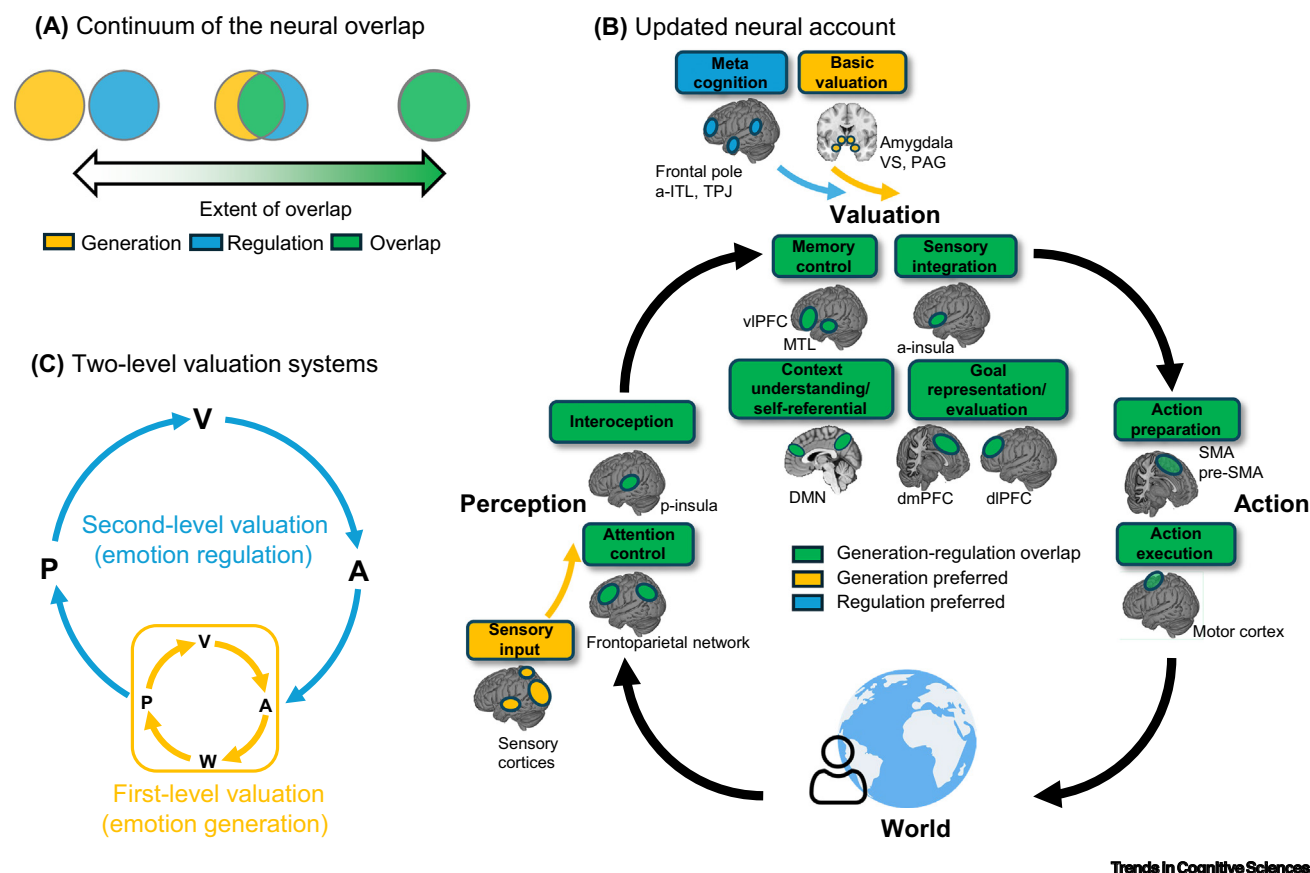


Figure 3. Brain bases of emotion generation and emotion regulation. (A) A continuum of the extent of overlap between the neural systems of emotion generation and emotion regulation. (B) An updated neural account of emotion generation and emotion regulation with overlapping and distinct neural substrates. We consider both emotion generation and emotion regulation as valuation systems in the form of ‘world–perception–valuation–action’ (WPVA) loops. Many subprocesses and their associated neural systems are shared between emotion generation and emotion regulation (green). There are also subprocesses that are preferentially involved in either emotion generation (yellow) or emotion regulation (blue). (C) The process model of emotion regulation [83] considers emotion generation as a first-level valuation system and emotion regulation as a second-level valuation system. Abbreviations: a-insula, anterior insula; a-ITL, anterior inferior temporal lobe; dlPFC, dorsolateral prefrontal cortex; DMN, default mode network; dmPFC, dorsomedial prefrontal cortex; MTL, medial temporal lobe; p-insula, posterior insula; PAG, periaqueductal gray; SMA, supplementary motor area; TPJ, temporoparietal junction; vIPFC, ventrolateral prefrontal cortex; VS, ventral striatum.

have little to no overlap. In the middle zone sit appraisal perspectives which hold that emotion results from situated meaning making, and that emotion generation and emotion regulation rely on meaning making in different ways and to different degrees [80]. On the other side (right in Figure 3A) stand various constructionist perspectives. Psychological constructionism theories (e.g., [81]) hold that it is difficult to distinguish emotion generation from emotion regulation, and social constructionism posits that separability is even less likely, as emotions are constructed in social context rather than ‘inside the head’ (e.g., [82]).

Considering the neuroimaging and neurostimulation evidence available to date, we propose an updated account, one that sits between these two ends of the continuum and is similar to appraisal models in the degree to which generation and regulation are separable. In this account, both emotion generation and emotion regulation may be thought of in terms of ‘world–perception–valuation–action’ (WPVA) loops (Figure 3B), value-construction processes that update iteratively during the course of experience based on sensory inputs, memories, and goals [83]. Like many brain processes, they unfold cyclically in the brain [84,85]. In this view there is a large set of overlapping

brain systems common to emotion generation and emotion regulation, but also a smaller set of brain systems preferentially supporting each of them in a context-dependent manner. This account is broadly consistent with findings on affective responses in non-human animals regarding the dual role of PFC in both pain generation and regulation, brain-wide neuronal activity during affective states, and variability in the neural circuitry involved in generating emotion of a particular category depending on the context (Box 1).

Regarding neural overlap, many subprocesses engaged by emotion generation and emotion regulation are shared at various stages in the WPVA valuation system (Figure 3B). During the ‘perception’ stage of both the emotion-generation valuation system and the emotion-regulation valuation system, the brain receives and processes exteroceptive sensations and interoceptive signals related to bodily sensations (preferentially associated with posterior insula and MCC) [86]. The brain allocates selective attention to the context-relevant features of these inputs (attention control, associated with the fronto-parietal network) [87]. During the ‘valuation’ stage of these valuation systems, the brain controls the retrieval of relevant autobiographical memory (memory control), integrates exteroceptive and interoceptive inputs (sensory integration), and conceptualizes the temporal–spatial context that situates the self (context understanding), associated with a network of the vPFC, vmPFC, anterior insula, and medial temporal lobe (MTL) [88,89]. Further, it implicitly or explicitly sets a goal state (goal representation), and compares the current state to the goal state (evaluation), associated with the dorsal PFC [90,91]. These valuation-related processes interactively and iteratively converge to form a conscious appraisal of one’s internal state, associated with medial parts of PFC [92,93]. During the ‘action’ stage, the brain makes (physiological and behavioral) action plans to reduce the discrepancy between the current state to the goal state and executes these action plans. This stage is preferentially associated with the pre-SMA/SMA and motor cortex [94]. All these subprocesses can be active in either emotion

Box 1. Affect generation and regulation in non-human animals

Research on basic affective processes in non-human animals can help to shed light on the neural basis of affective processes in humans. Such research is valuable because it enables the use of invasive and biochemical methods (e.g., optogenetics, neuropixels, viral tracing, calcium imaging) to observe and manipulate the brain at scales that are impossible in human research. Findings from recent animal research are congruent with the idea of considerable neural overlap between emotion generation and emotion regulation.

First, the frontal lobe seems to play a dual role in both perceiving and inhibiting pain. Although the PFC is often considered an inhibitor in the perception of pain [108,109], recent research reveals that it can also facilitate pain perception. A recent study [110] demonstrated in mice that neuronal engrams of long-term fear memory in the PFC enhanced pain experience later in life. When the prefrontal fear engrams were silenced, the chronically established pain could be reversed. Besides, recent animal research identified a frontal–insula pathway that induces and maintains pain hypersensitivity [111] and found that electrophysiological or optogenetic stimulation of the anterior cingulate cortex in this pathway facilitates pain perception [112].

Second, affective states emerge as a result of brain-wide neuronal activity. For example, one study [113] recorded thousands of neurons throughout the mice brain during thirst-motivated behavior. They discovered that when mice are thirsty, their brains exhibit widespread rather than local activation in response to cues of drinking behavior. Furthermore, optogenetically stimulating thirst-sensing neurons in a local brain region (hypothalamus) reinstated drinking behavior and brain-wide neuronal dynamics that encoded thirst. Their findings suggest that the brain states of thirst and possibly other affective states are manifested by widespread and interconnected neural activity throughout the brain, instead of activity confined to subcortical regions. Similar findings were found in the visual and motor domains [114].

Third, neural circuitry underpinning affective responses may vary depending on the context. Using the ‘fear’ paradigm in mice, researchers have found that whether a neural structure participates in a ‘fear’ response depends on the specific behavior or context. Exposing mice to threats, researchers [115] demonstrated that cells in a hypothalamic nucleus were activated during escape, but not other defensive behaviors like freezing and stretch-attending. In a related study, they showed that activity in this hypothalamic nucleus is necessary for context-specific escape but not panic-related escape caused by perceived asphyxiation [116].

generation or emotion regulation, depending on whether the target of interest is a stimulus that triggers emotion (emotion generation) or the emotion itself (emotion regulation). It is important to note that, due to the interactive nature of emotion generation and emotion regulation, the same neural structures may be engaged by each process at different time points, suggesting a possibility of temporal separability. However, considering available evidence, our focus here is spatial rather than temporal neural separability.

Although there are overlaps, some subprocesses typically preferentially subserve either emotion generation or emotion regulation (Figure 3B). Emotion generation is typically more strongly subserved by perception of external stimuli (associated with visual, auditory, and somatosensory cortices) [95] and basic valuation processes directly linked to survival and reward (associated with subcortical nuclei such as the amygdala, VS/NAcc, and PAG) [96–98]. Note that these generation-preferred regions are not necessarily susceptible to influences of all emotion regulation strategies, but could be influenced by some regulation practices [77,99]. However, emotion regulation (compared with emotion generation) typically requires meta-cognition of one's own emotions and higher-level goal representation at the valuation stage [77,100,101], which are potentially subserved by regulation-preferred brain structures found in fMRI studies, such as the frontal pole and TPJ [77,100,101]. These regulation-preferred regions are spatially adjacent to regions that are commonly activated in both emotion generation and emotion regulation (generation–regulation overlap), but are much smaller in volume and are thought to be situated at a higher level in the cognitive hierarchy.

Thus, in our view, some cognitive processes – and their associated neural circuits – are preferentially involved in either emotion generation or regulation, but the degree of separability is context-dependent. The empirical findings of generation- and regulation-preferred regions discussed herein were based on a simple image-viewing paradigm. In more complex contexts – such as reflecting on being wronged by a colleague, leading to feelings of anger – these ‘regulation-preferred’ regions might also be engaged due to the meta-cognitive processing involved in evaluating the situation and its emotional implications [65]. Here, we are not proposing a bright boundary between the emotion-generative and emotion-regulatory neural systems. Instead, we mean to suggest that certain subprocesses typically engaged in emotion regulation (e.g., meta-cognition) are less frequently involved in emotion generation, and vice versa. In scenarios where generating complex emotions requires meta-cognition or abstract evaluation, these regulation-preferred regions may also become involved in the generative process. In future work, connectivity studies may be particularly important in determining the context-dependent activation of these regions [37,102–105].

Our updated neural account is well aligned with the process model of emotion regulation [83], among other helpful theoretical frameworks [80,81]. The process model of emotion regulation describes emotion generation and emotion regulation as two conceptually separable but interactive WPVA valuation systems (Figure 3C). Emotion generation is instantiated as a first-level valuation system. In this system, the external or internal world (W) is perceived (P) and then evaluated (V) as good for me versus bad for me, and mental or physical actions (A) follow to address the gap between the perceived state and the desired state. Emotion regulation takes the first-level valuation system (i.e., emotion generation) as its target, and takes the form of a second-level valuation system, where its W is the generated emotion, its P is the perception of this emotion, its V is the evaluation of the emotion relative to a goal state, and its A represents an attempt to alter the emotion. Depending on the regulation strategy, this second-level A could act upon any of the four stages of the first-level WPVA valuation system (e.g., situational strategies that change the world; attentional strategies that change perception; cognitive strategies that change evaluation; and

response strategies that change the emotional responses). Although our discussion is largely focused on explicit emotion regulation, this framework also applies to implicit emotion regulation. When the representation of a goal state and the implementation of emotion regulation are outside one's awareness, the brain implicitly compares the perceived state with a goal state and implements some regulation strategy to reduce the gap. In this framework of interacting valuation systems, emotion generation and regulation should (i) share the same neural basis of the common scaffoldings of valuation systems, and (ii) display neural distinctions depending on the target, the goal state, and the specific actions in the valuation systems, matching our proposed account.

Concluding remarks

Our proposed neural account suggests that there are large spatial overlaps between the neural systems supporting emotion generation and the neural systems supporting emotion regulation. On this account, many of these neural systems subserve common subprocesses. Other neural systems preferentially subserve distinct subprocesses. A number of questions remain (see [Outstanding questions](#)), and to better understand the interplay among relevant brain systems, we suggest five directions for future research.

First, given the substantial neural overlap between emotion generation and emotion regulation, relevant research should avoid strong inferences from brain activation to function (i.e., generation or regulation) when interpreting results. For example, in a task involving emotion generation, stronger activation of dlPFC does not necessarily indicate stronger engagement of emotion regulation. Rather, brain activation should be interpreted with regard to comparisons between task events (e.g., regulate-emotional versus look-emotional to indicate emotion regulation). Relatedly, when studying emotion regulation, experiments involving brain observation and stimulation should always consider including a baseline condition of natural emotion generation (e.g., look-emotional condition) in order to better isolate the effects on emotion regulation. Otherwise, partial or even all effects could be due to impacts on emotion generation processes.

Second, most of the evidence supporting our neural account comes from studies using visual stimuli that induce negative emotions, using primarily downregulation via reappraisal, as the majority of studies testing these hypotheses have focused on such conditions. However, our ultimate goal is to examine whether this account generalizes across diverse circumstances. Therefore, future research involving different valences, alternative emotion regulation strategies, and various sensory modalities is encouraged to build upon and further validate this account.

Third, while we propose an account with shared subprocesses for both emotion generation and emotion regulation, this understanding has been derived largely from comparisons with other cognitive tasks. Future research should focus on experiments that directly test these specific subprocesses, with attention to the possibility that the engagement of particular common and distinct subprocesses may be determined by contextual factors such as the type/intensity of emotion, as well as the type/timing of emotion regulation. Novel task paradigms can be designed to better dissociate the specific role of a neural circuit in emotion generation versus emotion regulation. For example, different types of stimuli can be used to differentiate between emotion generation or emotion regulation in self versus non-self, exteroceptive versus interoceptive, social versus non-social, and memory-intensive versus non-memory-intensive contexts.

Fourth, as the role of particular subprocesses becomes clearer, interventions (e.g., psychosocial, brain stimulation, neurofeedback) should be designed to target them more precisely, potentially offering more effective treatments for affective disorders by addressing the most relevant components for a given person in the proposed account. Particularly, with the advance of spatially

Outstanding questions

To what extent are the brain systems that support emotion generation and emotion regulation context-specific? Current emotion induction paradigms have predominantly relied on static images to elicit emotional responses. More complex paradigms – such as those requiring extensive stimulus evaluation – could further blur the distinction between generation and regulation.

How do brain systems supporting emotion generation and emotion regulation operate in everyday life? Experiments typically use event-related tasks in which participants are directed either to respond naturally to emotional stimuli or to use specific forms of emotion regulation. In naturalistic settings, the boundary between emotion generation and emotion regulation may be less clearly defined than in the laboratory.

How does training influence the brain bases of emotion generation and emotion regulation? In cognitive behavioral therapy, regulation strategies are usually practiced over a period of months. Might the corresponding brain bases change through learning?

How generalizable are neural insights gleaned from studies on emotion generation and emotion regulation to other forms of self-control and other forms of affect? While emotion is central to mental well-being, effectively managing stress, addiction, pain, and other affective states also contributes to psychological health. If similar brain processes emerge across these varied domains, a single framework could potentially encompass a broad spectrum of affective phenomena central to mental well-being.

How does neural separability between emotion generation and emotion regulation relate to separability at a psychological level? Could emotion generation and emotion regulation comprise different instantiations of the same set of psychological processes? How separable are their neural bases at finer spatial and temporal scales than we can currently measure?

precise brain stimulation techniques, such as intracranial deep brain stimulation [106] and transcranial focused ultrasound stimulation [107], future research should consider stimulating brain regions expected to be regulation-preferred in a particular context to maximize the effect of emotion regulation enhancement.

Finally, both emotion generation and emotion regulation unfold over time. However, most research reviewed here relies heavily on fMRI, which lacks the temporal resolution necessary to capture the dynamic process within these brain regions. Future studies should combine fMRI with imaging protocols that offer higher temporal resolution, such as magnetoencephalography (MEG) or intracranial electroencephalography (iEEG). Additionally, within fMRI, more flexible modeling approaches with flexible basis functions (e.g., splines and finite impulse model) can be used to capture more complex temporal dynamics of the BOLD response.

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Declaration of interests

The authors declare no competing interests.

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